

However, the eradicated interactions can cause an increase in computation speeds which can be adjusted by changing the number of neurons.

Classification-Prediction ANN: Commonly used in data mining projects, classification neural networks are used to find particular patterns and group them into specific domains.

These patterns are then fed back to the system which produces newer classifications.

Radial Basis Function: These compare the values from the centre and have two layers. Similar to the points on a circle, the Euclidean distance is measured between the points. If the data point lies inside the radial distance it is classified within that domain otherwise it is excluded.

RBFs are the staple tool used by electrical companies to fix power systems during power outages.

Convolutional Neural Network: They share their structure to the feed-forward neural networks where all layers have separate weights and biases. Electrical students would best know them for being used in signal and image processing for analog and digital devices.

What makes a neural network work

Here's a strange truth about neural network simulations. You're never likely to yield the same model even when working on the same data set a multiple number of times. Sounds confusing, but try to understand that while analysts may type in the same code with the same definitions and variables, it's ultimately the machine that has to learn about the data.

And every time the machine learns the data, the methods and ideas it uses to connect the input to the output will differ. This is why it's well suggested to always configure the accuracy of a neural network program for one sample run and to save the results before executing future simulations.

In academic softwares like SPSS and MATLAB, the user runs a neural network program and then records the input weights which happen to be the unknown values to be found. The user tweaks the training and testing ratios, changes the learning rates, updates some hidden layer functions and runs the program once again.

Having generated some weights already, the system will likely arrive at new weights faster, and with greater accuracy, as it already has some weights to work it, i.e., it has known or learnt that the new weights must be close to the old ones.